AC and Transient Analysis

A Formula Reference for Linear Circuits — Inductors, Capacitors, Impedance, Charge, Flux, Voltage & Current

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This reference collects the constitutive relations of passive elements, the governing equations of first- and second-order transients, and the phasor/impedance description of sinusoidal steady state. Throughout, the passive sign convention is used: current i enters the terminal at which the voltage v is taken to be positive. The imaginary unit is written j so as not to clash with current, and $\omega = 2\pi f$ denotes angular frequency.

1 Constitutive Relations of Passive Elements

The three ideal two-terminal elements are defined by how voltage and current relate at their terminals.

1.1 Resistor

A resistor obeys Ohm's law instantaneously; it stores no energy.

$$v(t) = R\,i(t), \qquad i(t) = \frac{v(t)}{R}, \qquad p(t) = v\,i = i^2 R = \frac{v^2}{R}.$$

1.2 Capacitor — charge and current

A capacitor stores energy in its electric field. Its defining variable is the stored charge q, proportional to the terminal voltage through the capacitance C:

$$q(t) = C\,v(t), \qquad i(t) = \frac{\mathrm{d}q}{\mathrm{d}t} = C\,\frac{\mathrm{d}v}{\mathrm{d}t}, \qquad v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,\mathrm{d}\tau.$$

With a known initial voltage $v(t_0)$,

$$v(t) = v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\tau)\,\mathrm{d}\tau.$$

The voltage across a capacitor cannot change instantaneously for a finite current, so v_C is a continuous state variable.

1.3 Inductor — flux and voltage

An inductor stores energy in its magnetic field. Its defining variable is the flux linkage λ, proportional to the current through the inductance L:

$$\lambda(t) = L\,i(t), \qquad v(t) = \frac{\mathrm{d}\lambda}{\mathrm{d}t} = L\,\frac{\mathrm{d}i}{\mathrm{d}t}, \qquad i(t) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,\mathrm{d}\tau.$$

With a known initial current $i(t_0)$,

$$i(t) = i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(\tau)\,\mathrm{d}\tau.$$

The current through an inductor cannot change instantaneously for a finite voltage, so i_L is a continuous state variable.

1.4 Stored energy

$$W_C(t) = \tfrac{1}{2}\,C\,v^2 = \frac{q^2}{2C}, \qquad W_L(t) = \tfrac{1}{2}\,L\,i^2 = \frac{\lambda^2}{2L}.$$

	Resistor	**Capacitor**	**Inductor**
v–i law	$v = Ri$	$i = C\,\mathrm{d}v/\mathrm{d}t$	$v = L\,\mathrm{d}i/\mathrm{d}t$
State variable	—	v_C (charge $q = Cv$)	i_L (flux $\lambda = Li$)
Stored energy	0	$\frac{1}{2}Cv^2$	$\frac{1}{2}Li^2$
Cannot jump	—	voltage	current

2 First-Order Transients

A circuit with a single energy-storage element reduces, via Thévenin reduction of the resistive part, to a first-order linear ODE. Let $x(t)$ be the state variable (v_C or i_L).

2.1 General first-order form

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{1}{\tau}\,x = \frac{1}{\tau}\,x_\infty, \qquad x(t) = x_\infty + \big[x(0^+) - x_\infty\big]\,\mathrm{e}^{-t/\tau}.$$

Here $x(0^+)$ is the value just after the switching instant, x_∞ is the final (steady-state) value, and τ is the time constant.

2.2 Time constants

$$\tau_{RC} = R_{\mathrm{th}}\,C, \qquad \tau_{RL} = \frac{L}{R_{\mathrm{th}}}.$$

R_{th} is the Thévenin resistance seen by the storage element with independent sources deactivated.

2.3 RC circuit: charging and discharging

For a capacitor charged through R from a source V_s (initially uncharged):

$$v_C(t) = V_s\Big(1 - \mathrm{e}^{-t/RC}\Big), \qquad i(t) = \frac{V_s}{R}\,\mathrm{e}^{-t/RC}.$$

For a capacitor discharging from V_0 through R:

$$v_C(t) = V_0\,\mathrm{e}^{-t/RC}, \qquad i(t) = -\frac{V_0}{R}\,\mathrm{e}^{-t/RC}.$$

2.4 RL circuit: energizing and de-energizing

For an inductor energized through R from V_s (initially zero current):

$$i_L(t) = \frac{V_s}{R}\left(1 - \mathrm{e}^{-Rt/L}\right), \qquad v_L(t) = V_s\,\mathrm{e}^{-Rt/L}.$$

For an inductor de-energizing from I_0 through R:

$$i_L(t) = I_0\,\mathrm{e}^{-Rt/L}, \qquad v_L(t) = -I_0 R\,\mathrm{e}^{-Rt/L}.$$

After 5τ the response is within 0.7% of its final value and is treated as settled.

3 Second-Order Transients (Series & Parallel RLC)

With both L and C present the natural response satisfies a second-order ODE. Writing the source-free equation for the state variable x:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\alpha\,\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2\,x = 0, \qquad s^2 + 2\alpha s + \omega_0^2 = 0, \qquad s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}.$$

3.1 Defining parameters

$$\omega_0 = \frac{1}{\sqrt{LC}}\ \text{(undamped natural freq.)}, \qquad \zeta = \frac{\alpha}{\omega_0}\ \text{(damping ratio)}, \qquad Q = \frac{1}{2\zeta}.$$

Quantity	**Series RLC**	**Parallel RLC**
Neper frequency α	$\frac{R}{2L}$	$\frac{1}{2RC}$
Resonant freq. ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor Q	$\frac{1}{R}\sqrt{\frac{L}{C}}$	$R\sqrt{\frac{C}{L}}$

3.2 Damping cases

- **Overdamped** ($\alpha > \omega_0$): $x(t) = A_1\mathrm{e}^{s_1t} + A_2\mathrm{e}^{s_2t}$.
- **Critically damped** ($\alpha = \omega_0$): $x(t) = (A_1 + A_2t)\,\mathrm{e}^{-\alpha t}$.
- **Underdamped** ($\alpha < \omega_0$): $x(t) = \mathrm{e}^{-\alpha t}\big(A_1\cos\omega_d t + A_2\sin\omega_d t\big)$, with damped frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$.

4 Sinusoidal Steady State and Phasors

A sinusoid $x(t) = X_m\cos(\omega t + \phi)$ is represented by the phasor $\mathbf{X} = X_m\,\mathrm{e}^{\mathrm{j}\phi} = X_m\angle\phi$, recovered through

$$x(t) = \mathrm{Re}\big\{\mathbf{X}\,\mathrm{e}^{\mathrm{j}\omega t}\big\}.$$

Time differentiation maps to multiplication by jω, and integration to division by jω:

$$\frac{\mathrm{d}}{\mathrm{d}t} \longleftrightarrow \mathrm{j}\omega, \qquad \int \mathrm{d}t \longleftrightarrow \frac{1}{\mathrm{j}\omega}.$$

4.1 Impedance and admittance

Impedance $Z = V/I$ generalizes resistance to the complex frequency domain; admittance is its reciprocal $Y = 1/Z = G + \mathrm{j}B$.

$$Z_R = R, \qquad Z_L = \mathrm{j}\omega L, \qquad Z_C = \frac{1}{\mathrm{j}\omega C} = -\frac{\mathrm{j}}{\omega C}.$$

In rectangular and polar form,

$$Z = R + \mathrm{j}X = |Z|\,\mathrm{e}^{\mathrm{j}\theta}, \qquad |Z| = \sqrt{R^2 + X^2}, \qquad \theta = \arctan\frac{X}{R},$$

where X is the reactance: $X_L = \omega L > 0$ for an inductor and $X_C = -1/(\omega C) < 0$ for a capacitor.

4.2 Series and parallel combination

$$Z_{\text{series}} = \sum_k Z_k, \qquad \frac{1}{Z_{\text{parallel}}} = \sum_k \frac{1}{Z_k}, \qquad Z_{1\|2} = \frac{Z_1 Z_2}{Z_1 + Z_2}.$$

4.3 Resonance

Series and parallel RLC circuits resonate when the reactive parts cancel, $\omega_0 L = 1/(\omega_0 C)$:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}, \qquad \text{bandwidth BW} = \frac{\omega_0}{Q}.$$

5 AC Power

For voltage and current phasors $\mathbf{V} = V_m\angle\theta_v$, $\mathbf{I} = I_m\angle\theta_i$, define the phase difference $\theta = \theta_v - \theta_i$ and RMS magnitudes $V_{\text{rms}} = V_m/\sqrt{2}$, $I_{\text{rms}} = I_m/\sqrt{2}$.

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\,\mathbf{I}^*_{\text{rms}} = P + \mathrm{j}Q, \qquad |\mathbf{S}| = V_{\text{rms}} I_{\text{rms}}.$$
$$P = V_{\text{rms}} I_{\text{rms}} \cos\theta \;\; \text{(real, W)}, \qquad Q = V_{\text{rms}} I_{\text{rms}} \sin\theta \;\; \text{(reactive, var)}.$$

The power factor is $\text{pf} = \cos\theta = P/|\mathbf{S}|$ — lagging for inductive loads ($\theta > 0$) and leading for capacitive loads ($\theta < 0$).